Notice of References Cited	Application/Control No. 14/459,046	Applicant(s)/Patent Under Reexamination KALIDINDI, SANYASI R.	
	Examiner PATRICIA A. LEITH	Art Unit 1655	Page 1 of 2

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-2010/0202980 A1	08-2010	Fogel; Dov	A61K36/605	424/48
*	B	US-2012/0034325 A1	02-2012	Vaidya; Shatrughna Prasad	A61K36/185	424/734
*	C	US-2005/0084547 A1	04-2005	Subbiah, Ven	A61K31/704	424/740
*	D	US-2005/0008710 A1	01-2005	Subbiah, Ven	A61K36/00	424/725
*	E	US-8,445,033 B2	05-2013	Vaidya; Shatrughna Prasad	A61K36/185	424/725
*	F	US-2010/0203178 A1	08-2010	Gupta; Suresh Kumar	A61K9/08	424/769
*	G	US-2007/0065456 A1	03-2007	Woods; Cindy J.P.	A61K31/045	424/195.17
*	H	US-2006/0280698 A1	12-2006	Gupta; Subhash	A61K8/97	424/50
*	I	US-2006/0147555 A1	07-2006	Tripathi; Yamini Bhushan	A61K36/185	424/725
*	J	US-2005/0227910 A1	10-2005	Yang, David J.	A61K9/0024	424/422
*	K	US-2004/0258775 A1	12-2004	Patel, Sureshchandra K.	A61K36/185	424/745
*	L	US-2002/0022060 A1	02-2002	Mathur, Beena	A61K36/185	424/725
*	M	US-9,155,768 B2	10-2015	Gutmann; Erik Edward	A61K45/06	1/1

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	Caraka Sa¼hit¢ - Edited & translated by P.V Sharma,Vol.-II : Chaukhamba Orientalia, Varanasi, Edn. 5th, 2000. [Time of origin 1000 BC-4th century] (one page print-out from Traditional Knowledge Digital Library (TKDL)
	V	Cheng et al. ANTIOXIDANT AND FREE RADICAL SCAVENGING ACTIVITIES OF TERMINALIA CHEBULA; Biol. Pharm. Bull. 26(9) 1331-1335 (2003).
	W	Gupta, P.C. BIOLOGICAL AND PHARMACOLOGICAL PROPERTIES OF TERMINALIA CHEBULA RETZ. (HARITAKI)- AN OVERVIEW; Int. J. of Pharmacy and Pharmaceutical Sciences; Vol. 4, Sup. 3, 2012, pp. 62-68.
	X	Latha et al. INFLUENCE OF TERMINALIA BELLERICA ROXB. FRUIT EXTRACTS ON BIOCHEMICAL PARAMTERS IN STREPTOZOTOCIN DIABETIC RATS; Int. J. of Pharmacology 6 (2): 89-96 (2010)

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

Notice of References Cited	Application/Control No. 14/459,046	Applicant(s)/Patent Under Reexamination KALIDINDI, SANYASI R.	
	Examiner PATRICIA A. LEITH	Art Unit 1655	Page 2 of 2

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
	A	US-				
	B	US-				
	C	US-				
	D	US-				
	E	US-				
	F	US-				
	G	US-				
	H	US-				
	I	US-				
	J	US-				
	K	US-				
	L	US-				
	M	US-				

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	Naik et al. FREE RADICAL SCAVENGING REACTIONS AND PHYTOCHEMCIAL ANALYSIS OF TRIPHALA, AN AYURVEDIC FORMULATION; Current Science, Vol. 90, No. 8, 25 April 2006.
	V	Shen et al. TANNINS AND RELATED COMPOUNDS ISOLATED FROM TERMINALIA BELLERICA; Planta Med 2010; 76-P56, 7 pages.
	W	Wikipedia: HYPERURICEMIA; Online, URL: < https://web.archive.org/web/20120203161708/http://en.wikipedia.org/wiki/Hyperuricemia> ,February 3, 2012, 3 pages.
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/459,046	08/13/2014	Sanyasi R. Kalidindi	NTR-109	2412

51079 7590 04/22/2016
AMIN TALATI & UPADHYE, LLC
100 South Wacker Drive
Suite 2000
CHICAGO, IL 60606

EXAMINER	
LEITH, PATRICIA A	
ART UNIT	**PAPER NUMBER**
1655	
NOTIFICATION DATE	**DELIVERY MODE**
04/22/2016	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

patent@amintalati.com
george@amintalati.com
brent@amintalati.com

Office Action Summary	**Application No.** 14/459,046	**Applicant(s)** KALIDINDI, SANYASI R.	
	Examiner PATRICIA A. LEITH	**Art Unit** 1655	**AIA (First Inventor to File) Status** Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE _3_ MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☒ Responsive to communication(s) filed on _4/7/2016_.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on ____.

2a)☐ This action is **FINAL**. 2b)☒ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5)☒ Claim(s) _1-22_ is/are pending in the application.

5a) Of the above claim(s) _3-6,8-11 and 13-22_ is/are withdrawn from consideration.

6)☐ Claim(s) ____ is/are allowed.

7)☒ Claim(s) _1, 2, 7 and 12_ is/are rejected.

8)☐ Claim(s) ____ is/are objected to.

9)☐ Claim(s) ____ are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on ____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. ____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☒ Notice of References Cited (PTO-892)

2) ☒ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _(3) attached hereto_.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date. ____ .

4) ☐ Other: ____.

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

DETAILED ACTION

Terminalia chebula may be referred to herein as 'TC.'

Terminalia bellerica (or more recently known as *Terminalia bellirica*) may be referred to herein as 'TB.'

Election/Restrictions

Applicant's election without traverse of Group I claims 1-6 and 7-11 as well as the species of a) *T. bellerica*, b) fruit c) aqueous extract (of the fruit) and d) gout, uricemia and hyperuricemia in the reply filed on April 7, 2016 is acknowledged.

Applicant further requests the rejoinder of Groups I and II based on the consideration that the Examiner indicated in the Restriction Requirement mailed January 7, 2006 that if Applicant elected one or more of gout, uricemia or hyperuricemia, that Groups I and II would be examined on their merits.

Applicant's request is granted based upon the election of species of gout, uricemia and hyperuricemia and therefore Groups I and II are rejoined for examination. Please note that if Applicant subsequently amends any claim to reflect a scope other than the elected species, certain claims may be withdrawn accordingly. For example, if Applicant amends claim 1 to require that the free radical induced disease is hypertension or diabetes for example (see page 4 of the Specification), but does not amend Group II accordingly, Group II would be continued to be examined considering that this species of hyperuricemia overlaps with the elected species and Group I would be withdrawn from consideration.

Claims 5, 6, 10, 11 and 18 as well as dependent claims thereof are based upon an unknown extract. The phrase ‘aqueous extract’ is known in the art as an essential equivalent to a ‘water extract’ *i.e.,* an extract obtained with water and whereby an ‘aqueous extract’ may also, as opposed to a strict ‘water extract’ may also comprise ionized components such as salts and buffers. While certain parameters may fluctuate in such an extraction procedure such as the temperature, pressure and amount of water/buffer/salt the addition of steps outside of these parameters will often not coincide with an ‘aqueous extract’ of the plant material. For example, if one extracts the TB plant with water and then fractionates such an extract using HPLC to collect a concentrated fraction, this fraction is not considered an ‘aqueous extract’ of the plant. Rather, this would be considered a ‘fraction’ of the aqueous extract of the plant.

The percentage of constituents and ED50 values as disclosed were calculated from extracts of TB *obtained from Natreon, Inc.* (Example 1). Applicant provides no indication of what these extracts were or how they were made. Moreover, Applicant indicates that optimization must occur in order to achieve certain percentages of endogenous components (*e.g.*, p. 17, Specification) but does not teach what those optimization parameters are in order to achieve the claimed percentages of components. Therefore, there is no evidence set forth within the Instant specification that the percentages of components or the ED50 values are associated with an *aqueous extract* of TB as elected.

The *in-vivo* study was carried out with an aqueous extract of TB fruits whereby the extract contained chebulinic acid and chebulagic acid, gallic and ellagic acid. However, there is no indication that this aqueous extract contains the percentages of constituents as claimed in the dependent claims.

Claims 3-6, 8-11 and 13-22, solely directed toward the non-elected TC extract are hereby withdrawn from consideration at this time.

Claims 1, 2, 7 and 12 were examined on their merits.

Claim Rejections - 35 USC § 102

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale or otherwise available to the public before the effective filing date of the claimed invention.

Claims 1, 2 and 12 are rejected under 35 U.S.C. 120 (a)(1) as being anticipated by Caraka Sa¼hit¢ - Edited & translated by P.V Sharma,Vol.-II : Chaukhamba Orientalia, Varanasi, Edn. 5th, 2000. [Time of origin 1000 BC-4th century] (one page print-out from Traditional Knowledge Digital Library (TKDL) (Reference referred to herein as 'Caraka') as evidenced by* Wikipedia: HYPERURICEMIA; Online, URL: < https://web.archive.org/web/20120203161708/http://en.wikipedia.org/wiki/Hyperuricemia> ,February 3, 2012, 3 pages.

Caraka discloses a decoction (water extract which is boiled and filtered to obtain a Kvatha (sometimes spelled 'Quatha' or 'kwatha' in Ayurvedic medicine) of TC and TB fruit along with Phyllanthus embilica fruit as a treatment for gout.

As evidenced by Wikipedia (archived to February 3, 2012) , gout is caused by hyperuricemia which is caused by high uric acid levels in the blood (uric acid in the blood being known as 'uricemia'). Thus, persons having gout are also concomitantly suffering from uricemia/hyperuricemia. Therefore, a treatment of gout would also be considered a treatment of uricemia/hyperuricemia since treatment of gout would necessarily need to lower the uric acid level of the blood/serum.

As evidenced by Applicant's Specification, the aqueous extract of TB fruit contains chebulinic acid [0086]. Therefore, the composition disclosed by Caraka will also inherently contain chebulinic acid.

It is noted that there is no requirement that a person of ordinary skill in the art would have recognized the inherent disclosure at the time of invention, but only that the subject matter is in fact inherent in the prior art reference. *Schering Corp. v. Geneva Pharm. Inc.*, 339 F.3d 1373, 1377, 67 USPQ2d 1664, 1668 (Fed. Cir. 2003) (rejecting the contention that inherent anticipation requires recognition by a person of ordinary skill in the art before the critical date and allowing expert testimony with respect to post-critical date clinical trials to show inherency); see also *Toro Co. v. Deere & Co.*, 355 F.3d 1313, 1320, 69 USPQ2d 1584, 1590 (Fed. Cir. 2004)("[T]he fact that a characteristic is a necessary feature or result of a prior-art embodiment (that is itself sufficiently described and enabled) is enough for inherent anticipation, even if that fact was unknown at the time of the prior invention."); *Atlas Powder Co. v. Ireco, Inc.*, 190 F.3d 1342, 1348-49 (Fed. Cir. 1999) ("Because sufficient aeration' was inherent in the prior art, it is irrelevant that the prior art did not recognize the key aspect of [the] invention…. An inherent structure, composition, or function is not necessarily known.")>; *SmithKline Beecham Corp. v. Apotex Corp.*, 403 F.3d 1331, 1343-44, 74 USPQ2d 1398, 1406-07 (Fed. Cir. 2005) (holding that a prior art patent to an anhydrous form of a compound "inherently" anticipated the claimed hemihydrate form of the compound because practicing the process in the prior art to manufacture the anhydrous compound "inherently results in at least trace amounts of" the claimed hemihydrate even if the prior art did not discuss or recognize the hemihydrate) (See MPEP 2112).

Claim 1 indicates, in-part 'wherein xanthine oxidase is inhibited' and claim 12 indicates in-part '…wherein serum uric acid is lowered.'

Applicant is directed to MPEP § 2111.04:

> The determination of whether each of these clauses is a limitation in a claim depends on the specific facts of the case. *In Hoffer v. Microsoft Corp.*, 405 F.3d 1326, 1329, 74 USPQ2d 1481, 1483 (Fed. Cir. 2005), the court held that when a "whereby' clause states a condition that is material to patentability, it cannot be ignored in order to change the substance of the invention." Id. However, the court noted (quoting *Minton v. Nat 'l Ass 'n of Securities Dealers, Inc.,* 336 F.3d 1373, 1381, 67 USPQ2d 1614, 1620 (Fed. Cir. 2003)) that a ***"whereby clause in a method claim is not given weight when it simply expresses the intended result of a process step positively recited.'"*** Id.< (emphasis added).

Hence, it follows from *Minton v. Nat 'l Ass 'n of Securities Dealers, Inc.*, 336 F.3d 1373, 1381, 67 USPQ2d 1614, 1620 (Fed. Cir. 2003) that the language '…wherein xanthine oxidase is inhibited' and '…wherein serum uric acid is lowered' hold no patentable weight because these statements do not further limit the actual method steps of the respective claims.

The issue regarding stating what occurs as a *result* of a positively-recited method step and providing this statement patentable weight is that one would need to essentially guess how Applicant means to limit the positively-recited method steps by stating a result of the method. Herein lies the reason why these clauses are not given patentable weight.

Therefore, treatment of gout (and thus necessarily uricemia/hyperuricemia) with an aqueous extract of TB fruit was known in the art and is therefore anticipated by Cheng et al.

Claim 7 is rejected under 35 U.S.C. 120 (a)(1) as being anticipated by Naik et al. FREE RADICAL SCAVENGING REACTIONS AND PHYTOCHEMCIAL ANALYSIS OF TRIPHALA, AN AYURVEDIC FORMULATION; Current Science, Vol. 90, No. 8, 25 April 2006.

Naik et al. teach that Triphala is an Ayurvedic formulation made by extracting fruits of three plants, one of the fruits originating from TB and whereby the fruits are extracted with water (see protocol for making Triphala: col. 2, page 1100). Naik et al. indicate that Triphala is "…commonly prescribed by most healthcare practitioners in India" as it is known for possessing many therapeutic properties as anti-viral, anti-bacterial, detoxification and improving blood circulation/blood pressure (p. 1100, col. 1).

Naik et al. demonstrate that Triphala inhibits Xanthine Oxidase (entire reference, especially Figure 2).

Therefore, claim 7 is anticipated by Naik et al. because xanthine oxidase would have necessarily been inhibited in the persons taking Triphala prior to the document published by Naik et al. as evidenced by Naik et al. Please note that claim 7 is directed toward inhibition of any amount of inhibition of XO and therefore, administration of Triphala need have only inhibited one mole of XO. Since Triphala (water extract thereof) has XO inhibiting capabilities, this composition would have inhibited at least one mole of XO upon *in-vivo* administration.

It is noted that there is no requirement that a person of ordinary skill in the art would have recognized the inherent disclosure at the time of invention, but only that the subject matter is in fact inherent in the prior art reference. *Schering Corp. v. Geneva Pharm. Inc.*, 339 F.3d 1373, 1377, 67 USPQ2d 1664, 1668 (Fed. Cir. 2003) (rejecting the contention that inherent anticipation requires recognition by a person of ordinary skill in the art before the critical date and allowing expert testimony with respect to post-critical date clinical trials to show inherency); see also *Toro Co. v. Deere & Co.*, 355 F.3d 1313, 1320, 69 USPQ2d 1584, 1590 (Fed. Cir. 2004)("[T]he fact that a characteristic is a necessary feature or result of a prior-art embodiment (that is itself sufficiently described and enabled) is enough for inherent anticipation, even if that fact was unknown at the time of the prior invention."); *Atlas Powder Co. v. Ireco, Inc.*, 190 F.3d 1342, 1348-49 (Fed. Cir. 1999) ("Because sufficient aeration' was inherent in the prior art, it is irrelevant that the prior art did not recognize the key aspect of [the] invention.... An inherent structure, composition, or function is not necessarily known.")>; *SmithKline Beecham Corp. v. Apotex Corp.*, 403 F.3d 1331, 1343-44, 74 USPQ2d 1398, 1406-07 (Fed. Cir. 2005) (holding that a prior art patent to an anhydrous form of a compound "inherently" anticipated the claimed hemihydrate form of the compound because practicing the process in the prior art to manufacture the anhydrous compound "inherently results in at least trace amounts of" the claimed hemihydrate even if the prior art did not discuss or recognize the hemihydrate) (See MPEP 2112).

Conclusion

No claims are allowed.

*This reference is cited merely to relay an inherent/intrinsic property and is not used in the basis for rejection *per se*.

The prior art made of record on the accompanying PTO-892 form which is not relied upon is considered pertinent to Applicant's disclosure.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to PATRICIA A. LEITH whose telephone number is (571)272-0968. The examiner can normally be reached on Monday - Friday 9:00am-5:30pm.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Terry McKelvey can be reached on (571) 272-0775. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see http://pair-direct.uspto.gov. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

PATRICIA A LEITH
Primary Examiner
Art Unit 1655

/PATRICIA A LEITH/
Primary Examiner, Art Unit 1655
April 15, 2016

UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/459,046	08/13/2014	Sanyasi R. Kalidindi	NTR-109	2412

51079 7590 03/08/2017
AMIN TALATI UPADHYE LLP
100 South Wacker Drive
Suite 2000
CHICAGO, IL 60606

EXAMINER	
KOSAR, AARON J	

ART UNIT	PAPER NUMBER
1655	

NOTIFICATION DATE	DELIVERY MODE
03/08/2017	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

patent@amintalati.com
george@amintalati.com
brent@amintalati.com

Office Action Summary	**Application No.** 14/459,046	**Applicant(s)** KALIDINDI, SANYASI R.	
	Examiner AARON J. KOSAR	**Art Unit** 1655	**AIA (First Inventor to File) Status** Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE *3* MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☒ Responsive to communication(s) filed on *10/24/2016*.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☒ This action is **FINAL**. 2b)☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5)☒ Claim(s) *1-22* is/are pending in the application.
5a) Of the above claim(s) *3-6,8-11 and 13-22* is/are withdrawn from consideration.
6)☐ Claim(s) _____ is/are allowed.
7)☒ Claim(s) *1, 2, 7 and 12* is/are rejected.
8)☐ Claim(s) _____ is/are objected to.
9)☐ Claim(s) _____ are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

10)☒ The specification is objected to by the Examiner.
11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).
Certified copies:
a)☐ All b)☐ Some** c)☐ None of the:
1.☐ Certified copies of the priority documents have been received.
2.☐ Certified copies of the priority documents have been received in Application No. _____.
3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☒ Notice of References Cited (PTO-892)
2) ☒ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date *(10/24/2016*.
3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date. _____ .
4) ☐ Other: _____.

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA. Please note the assigned Examiner information has been updated as indicated at the end of this office action.

DETAILED ACTION

Response to Amendment

Applicant's amendment and argument filed 24 October 2016, in response to the non-final rejection, are acknowledged and have been fully considered. Any previous rejection or objection not mentioned herein is withdrawn.

<u>Note regarding Election/Restriction</u>: Applicant elected without traverse of Group I claims 1-6 and 7-11 as well as the species of: (1) gout, uricemia and hyperuricemia and (2-3) *T. bellerica* fruit aqueous extract. Additionally, as indicated in the previous office action, claim 12 (within group II) remains rejoined and examined herewith. Claims 1-22 are pending of which claims 3-6, 8-11 and 13-22 remain withdrawn.

Pending claims 1, 2, 7, and 12 have been examined on the merits.

Objections - Claims, Specification

Claims 1, 7, and 12 are objected to because of the following informalities:

In each of claims 1, 7, and 12, the term "*bellerica*" is considered a commonly-used misnomer in the art. The correct taxonomic species term should be recited as --*bellirica*-- .

The term “bellerica” should also be amended in each instant in the specification to recite --*bellirica*--. Additionally, at paragraph [0006], the phrase should be amended to emphasize the dominance of *T. bellirica*, for example reciting similar to the following: --*Terminalia bellirica* (Gaertn.) Roxb. (a.k.a. as *Terminalia bellerica*) ("TB")--.

Appropriate correction is required.

Claim Rejections - 35 USC § 102

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

A person shall be entitled to a patent unless –

(a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale or otherwise available to the public before the effective filing date of the claimed invention.

Claims 1, 2, and 12 stand as rejected under 35 U.S.C. 102 (a)(1) as being anticipated by Caraka Samhita (Sharma (Ed. & transl), vol I-II: Chaukhamba Orientalia, Varanasi, 5th Edn., 2000. [Time of origin 1000 BC-4th century] (Trad. Know. Dig. Lib. (TKDL), 1 pg.; herein “Caraka”) as evidenced by Wikipedia: Hyperuricemia (Online, URL: < https://web.archive.org/web/20120203161708/http://en.wikipedia.org/wiki/Hyperuricemia> , 3 Feb 2012, 3 pp., herein “Wiki (I)”)

Caraka discloses a decoction (water extract which is boiled and filtered to obtain a kvatha, (Ayurvedic medicine a.k.a.: quatha, kwatha), of *T. chebula* (TC) and *T. bellirica* (TB) fruit along with *Phyllanthus embilica* fruit as a treatment for gout.

As evidenced by Wiki (I) (archived to February 3, 2012), gout is caused by hyperuricemia which is caused by high uric acid levels in the blood (uric acid in the blood being known as 'uricemia'). Thus, where Caraka may be silent with respect to the underlying biochemical effect of administering the decoction/aqueous extract, or specifically recite a “hyperuricemia” or “xanthine oxidase” (XO) enzyme activity thereof, one of skill would have recognized that one suffering from gout are also concomitantly suffering from uricemia/hyperuricemia and that a treatment of gout would also be considered a treatment of uricemia/hyperuricemia/xanthine oxidase since treatment of gout would necessarily need to lower the uric acid level of the blood/serum (broadly an XO activity).

As evidenced by Applicant's Specification, the aqueous extract of TB fruit contains chebulinic acid [0086]. Therefore, the TB fruit comprising aqueous extract composition disclosed by Caraka will also inherently contain chebulinic acid. It is noted that there is no requirement that a person of ordinary skill in the art would have recognized the inherent disclosure at the time of invention, but only that the subject matter is in fact inherent in the prior art reference. *Schering Corp. v. Geneva Pharm. Inc*., 339 F.3d 1373, 1377, 67 USPQ2d 1664, 1668 (Fed. Cir. 2003) (rejecting the contention that inherent anticipation requires recognition by a person of ordinary skill in the art before the critical date and allowing expert testimony with respect to post-critical date clinical trials to show inherency); see also *Toro Co. v. Deere & Co.*, 355 F.3d 1313, 1320, 69 USPQ2d 1584, 1590 (Fed. Cir. 2004)(“[T]he fact that a characteristic is a necessary feature or result of a prior-art embodiment (that is itself sufficiently described and enabled) is enough for inherent anticipation, even if that fact was unknown at the time of the prior invention."); *Atlas Powder Co. v. Ireco, Inc*., 190 F.3d 1342, 1348-49 (Fed. Cir. 1999) ("Because sufficient aeration' was inherent in the prior art, it is irrelevant that the prior art did not recognize the key aspect of [the] invention…. An inherent structure, composition, or function is not necessarily known.")>; *SmithKline Beecham Corp. v. Apotex Corp*., 403 F.3d 1331, 1343-44, 74 USPQ2d 1398, 1406-07 (Fed. Cir. 2005) (holding that a prior art patent to an anhydrous form of a compound "inherently" anticipated the claimed hemihydrate form of the compound because practicing the process in the prior art to manufacture the anhydrous compound "inherently results in at least trace amounts of" the claimed hemihydrate even if the prior art did not discuss or recognize the hemihydrate) (See MPEP 2112).

Claim 1 indicates, in-part 'wherein xanthine oxidase is inhibited' and claim 12 indicates in-part '…wherein serum uric acid is lowered'. However, MPEP § 2111.04 states:

"The determination of whether each of these clauses is a limitation in a claim depends on the specific facts of the case. In Hoffer v. Microsoft Corp., 405 F.3d 1326, 1329, 74 USPQ2d 1481, 1483 (Fed. Cir. 2005), the court held that when a "whereby' clause states a condition that is material to patentability, it cannot be ignored in order to change the substance of the invention." Id. However, the court noted (quoting Minton v. Nat 'l Ass 'n of Securities Dealers, Inc., 336 F.3d 1373, 1381, 67 USPQ2d 1614, 1620 (Fed. Cir. 2003)) that a ***"whereby clause in a method claim is not given weight when it simply expresses the intended result of a process step positively recited.'"*** *Id.< (emphasis added).*

Hence, it follows from *Minton v. Nat 'l Ass 'n of Securities Dealers, Inc*., 336 F.3d 1373, 1381, 67 USPQ2d 1614, 1620 (Fed. Cir. 2003) that the language "…wherein xanthine oxidase is inhibited" and "…wherein serum uric acid is lowered" hold no patentable weight because these statements do not further limit the actual method steps of the respective claims (do not require more, or materially identify a novel difference, from the composition of the prior art.

Additionally, where the prior art and the instant claims administer the same TB fruit aqueous extract to the same target (gout-affected patient having XO enzyme and pathogenic uric acid levels), then the same effect would have been intrinsically present, and especially in the absence of objective evidence to the contrary.

The issue regarding stating what occurs as a *result* of a positively-recited method step and providing this statement patentable weight is that one would need to essentially guess how Applicant means to limit the positively-recited method steps by stating a result of the method. Herein lies the reason why these clauses are not given patentable weight.

Therefore, treatment of gout (and thus necessarily uricemia/hyperuricemia) with an aqueous extract of TB fruit was known in the art and is therefore anticipated by Cheng et al.

Response to Arguments

In response to applicant's argument (see declaration at III(A) for example and at the corresponding arguments in the Remarks) that the cited references, that "triphala is not disclosed by Caraka to treat vatarakta, or gout" (see 1.132 declaration page 7 at #23-26), a recitation of the intended use of the claimed invention must result in a structural difference between the claimed invention and the prior art in order to patentably distinguish the claimed invention from the prior art. If the prior art structure is capable of performing the intended use, then it meets the claim.

Accordingly, in response to Applicant's arguments, as evidenced by Naik et al ("Free radical scavenging reactions and phytochemical analysis of triphala, an ayurvedic formulation" Current Science (2006), 90(8), 1100-1105 CODEN: CUSCAM; ISSN: 0011-3891; Abstract only), "triphala was tested for superoxide radical scavenging activity using xanthine and xanthine oxidase assay, it was observed that in addition to reacting with superoxide radical, it also inhibited uric acid formation, indicative of xanthine oxidase enzyme [XO] inhibitory activity." Although not specifically reciting gout, uric acid and XO activity within the context of being an "ayurvedic formulation" and as symptoms of gout, are also broadly and reasonably considered to effect a gout treatment as the triphala composition would intrinsically provide the aforementioned uric acid and XO inhibitory (gout treating) effects. Also the claims are drawn to merely an "extract" that "comprises" *T. bellirica* (TB), and thus also remain inclusive of the extracting processes and products thereof of the prior art, including administering products containing materials in addition to TB such as triphala which as evidenced by Naik as discussed above provides a TB-comprising composition effecting XO activity and uric acid formation inhibitions and free (free)radical scavenging (Naik at abstract) Please note, in so far as the composition contains any amount of TB extract including trace amounts or deactivated (applicant terms "destroyed"), since the material may "comprise" additional components, then only the total composition effect is considered necessary, i.e. as long as *the composition as a whole* has the degree of the "therapeutic effect" to some degree of gout/xo-inhibition/anti-uric acid effect(s), then the administration of the composition to provide such an effect meets the claims.

In response to applicant's arguments that heating at elevated temperatures (80 °C, e.g. Appendix A remarks, through #11) would provide a reduced/trace/destroyed amount of TB active agent, this is not found to be persuasive, because applicant is arguing features (i.e. temperature(s) and amounts of chebulagic/chebulinic acids, LMWHT's; method extraction steps excluding the prior art process and compositions) which are not recited in the pending claims. Additionally, evidence of *isolated* TB agents is not found to be persuasive, because the claims broadly remain inclusive also of compositions comprising active and/or inactive TB extract amounts and additional materials whereby the "composition" as a whole provides the claimed effect. Accordingly (in addition to the remarks over #23-26 as argued above), regarding remark #28 that "Kvatha would have no TB active constituents left after boiling" this is not found to be persuasive, because the *composition* would still have a gout treating effect and would still contain TB extract, and further it is noted that the claims do not require amounts of chebulinic/chebulagic acid/LMWHT or correlate such materials to the *composition* activity in such a way to preclude the prior art composition (i.e. remain inclusive of the prior art).

In response to Applicants arguments that certain of the claims by Caraka are mythical or impossibilities, this is not found to be persuasive, because such motivations and non-functional embodiments (i.e. portions of the reference directed to for example "invincib[ility]", remarks through #31) do not negate that Caraka elsewhere teaches administering TB-containing compositions (and the recited and unrecited intrinsic effects that would have been present in each composition).

In response to Applicant's arguments about the "percentages of constituents in the dependent claims" (#16), this is not found to be persuasive, because the grounds are not directed to limitations of the pending, examined claims, but to claims that remain withdrawn.

In response to Applicant's argument that hyperuricemia can be treated in the absence of gout or symptoms thereof (#32), this is not found to be persuasive, because the therapeutic effect as claimed merely requires that any degree of xanthine oxidase (XO) inhibition (or administering any amount of TB extract which is broadly "therapeutically effective").

Claim 7 stands as rejected under 35 U.S.C. 102 (a)(1) as being anticipated by Naik et al. ("Free Radical Scavenging Reactions and Phytochemical Analysis of Triphala, an Ayurvedic Formulation"; Current Science, 25 April 2006, 90(8), 1100-1105)

Naik et al. teach that Triphala is an Ayurvedic formulation made by extracting fruits of three plants, one of the fruits originating from TB fruit and whereby the fruits are extracted with water (see protocol for making Triphala: col. 2, page 1100). Naik et al. indicate that Triphala is "…commonly prescribed by most healthcare practitioners in India" as it is known for possessing many therapeutic properties as anti-viral, anti-bacterial, detoxification and improving blood circulation/blood pressure (p. 1100, col. 1). Naik et al. also demonstrate that Triphala inhibits Xanthine Oxidase (entire reference, especially Figure 2). Therefore, claim 7 is anticipated by Naik et al. because xanthine oxidase would have necessarily been inhibited in the persons taking Triphala prior to the document published by Naik et al. as evidenced by Naik et al. Please note that claim 7 is directed toward any inhibition amount of XO and, therefore, administration of Triphala need have only inhibited one molecule of XO. Since Triphala (water extract thereof) has XO inhibiting capabilities, this composition would have inhibited at least one mole of XO upon *in-vivo* administration.

It is noted that there is no requirement that a person of ordinary skill in the art would have recognized the inherent disclosure at the time of invention, but only that the subject matter is in fact inherent in the prior art reference. *Schering Corp. v. Geneva Pharm. Inc.*, 339 F.3d 1373, 1377, 67 USPQ2d 1664, 1668 (Fed. Cir. 2003) (rejecting the contention that inherent anticipation requires recognition by a person of ordinary skill in the art before the critical date and allowing expert testimony with respect to post-critical date clinical trials to show inherency); see also *Toro Co. v. Deere & Co.*, 355 F.3d 1313, 1320, 69 USPQ2d 1584, 1590 (Fed. Cir. 2004)("[T]he fact that a characteristic is a necessary feature or result of a prior-art embodiment (that is itself sufficiently described and enabled) is enough for inherent anticipation, even if that fact was unknown at the time of the prior invention."); *Atlas Powder Co. v. Ireco, Inc.*, 190 F.3d 1342, 1348-49 (Fed. Cir. 1999) ("Because sufficient aeration" was inherent in the prior art, it is irrelevant that the prior art did not recognize the key aspect of [the] invention…. An inherent structure, composition, or function is not necessarily known.")>; *SmithKline Beecham Corp. v. Apotex Corp.*, 403 F.3d 1331, 1343-44, 74 USPQ2d 1398, 1406-07 (Fed. Cir. 2005) (holding that a prior art patent to an anhydrous form of a compound "inherently" anticipated the claimed hemihydrate form of the compound because practicing the process in the prior art to manufacture the anhydrous compound "inherently results in at least trace amounts of" the claimed hemihydrate even if the prior art did not discuss or recognize the hemihydrate) (See MPEP 2112).

Response to Arguments

Applicant has argued (and provided in the declaration at III(B)) that Naik teaches TB as "one of three" triphala components which does not correlate the XO inhibition activity of the composition to TB, and argues that as such that TC and TB in the composition "is highly likely...both *T. chebula* and *T. bellirica* were destroyed or rendered inert" by Naik's process.

In response to applicant's argument that the references fail to show these certain features of applicant's invention, it is noted that the features upon which applicant relies are not recited in the rejected claim(s). The claims merely require that TB be present in the composition, which triphala comprises an amount of a TB aqueous extract and that the composition as a whole contain a xanthine oxidase inhibitory effect, which would have been intrinsic to the composition and which functions to inhibit XO (fig 2 and at p.1102 left col. "triphala…has XO enzyme inhibitory activity" for example). Although the claims are interpreted in light of the specification, limitations from the specification are not read into the claims. See *In re Van Geuns*, 988 F.2d 1181, 26 USPQ2d 1057 (Fed. Cir. 1993). Additionally, it is noted that the TB extract is merely identified as an "extract" but does not expressly recite the active components present in the extract or correlate the claimed composition function to component(s) present therein (may comprise additional materials providing the function). Also, specifically in response to applicant's "one of three" component argument, this is not found to be persuasive, because the claims and triphala each "comprise" TB extract and embrace the additional unrecited materials (remain inclusive of a decoction and/or triphala composition).

THIS ACTION IS MADE FINAL. Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the mailing date of this final action.

Conclusion

No claims are allowed.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to AARON J. KOSAR whose telephone number is (571)270-3054. The examiner can normally be reached on Monday-Friday, 9:00AM-5:00PM, in general, EST.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Terry McKelvey can be reached on (571) 272-0775. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see http://pair-direct.uspto.gov. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/AARON J KOSAR/
Examiner, Art Unit 1655

/Christopher R Tate/
Primary Examiner, Art Unit 1655